Day 5: RAG (Retrieval-Augmented Generation) – Interview Q&A

Sagar DS · November 2026

RAG in one line: find the right information first, add it to the prompt, then let the LLM answer from it.

Memory trick: RAG gives the model a book to read. Fine-tuning changes the model's brain.

1. What do R, A and G stand for, and what does each step do?

Letter	Step	What it does
R	**Retrieval**	Find the most relevant pieces of information for the question
A	**Augmented**	Add those pieces to the prompt, together with the question and rules
G	**Generation**	The LLM writes the answer using that added information

How retrieval works, step by step

1. The question is converted into an **embedding**, using the **same** embedding model that was used for the documents.
2. The vector database compares it with the stored embeddings of the **document chunks** (one per section, not whole documents).
3. It returns the **top-k closest chunks** (for example the top 3), with their **text** and **source**.

Example: "Server is out of space" → **R:** find the disk-full runbook sections → **A:** put them in the prompt → **G:** the LLM writes the steps from them.

Interview line: "RAG retrieves relevant documents, augments the prompt with them, and the LLM generates an answer grounded in that content. In retrieval, the question is embedded with the same model as the documents, and the vector database returns the top-k closest chunks with their text and source."

2. Why does a plain LLM hallucinate about a company's own documents?

Because it **has never seen them**. An LLM only knows what was in its training data: general, public information up to a certain date. It knows nothing about private or recent documents. When asked, it does not say "I don't know"; it **predicts a plausible-sounding answer**, which can be invented.

Example: asked about a made-up command such as `diskclean --deep`, a model may confidently describe what it does.

Interview line: "LLMs generate plausible text from their training data. They don't have private or recent information, so without retrieval they fill the gap with guesses."

3. Why are citations important?

Reason	Meaning
Verification	The user can open the source and check the answer
Trust	Shows the answer comes from real documents, not invention
Debugging	If the answer is wrong, you know which document to fix
Accountability	Important in regulated environments such as finance and healthcare

Interview line: "Citations make RAG answers verifiable. Users can check the source, trust increases, and wrong answers can be traced back to the document that caused them."

4. What should the assistant say when the answer isn't in the documents, and why?

It should clearly say so, for example **"Not found in the knowledge base"**, instead of guessing.

Why: a confident wrong answer is **more dangerous** than no answer. In operations, someone might run an invented command on a production system. "Not found" tells the user to escalate or check elsewhere.

Interview line: "I instruct the model to answer only from the retrieved context and to say 'not found' otherwise, because in production a confident wrong answer is worse than no answer."

5. RAG or fine-tuning for runbooks, and why?

RAG.

	RAG	Fine-tuning
Documents change often	Edit the document, re-index in seconds	Retrain the model each time
Shows the source	Yes, with citations	No
Cost and effort	Low	High: time, hardware, expertise
Good for	**Facts and knowledge**	**Style or behaviour**: tone, format, specialised tasks

Interview line: "For changing factual knowledge like runbooks, RAG is better: it's cheap to update, shows sources and reduces hallucination. Fine-tuning suits teaching a model a style or behaviour, not keeping facts current."

Quick revision

Concept	One line
RAG	Retrieve → add to prompt → generate
Grounding	Answer only from the retrieved text
Citations	Always name the source document
"Not found"	Better than a confident wrong answer
Retrieval failure	Wrong chunks found, so the answer is wrong
Generation failure	Right chunks found, but the LLM ignores or misreads them
Golden rule	RAG is only as good as its retrieval